

Composition Pair A1.03 + B1.14 — The Resolution Pair Where Conflict-as-First-Class Registers Conflicts as Governance-Demanding Artifacts Requiring Resolution, and Directed Selection Provides the Governed DNA Modification Mechanism Through Which Registered Conflicts Are Addressed, Together Forming the Conflict Visibility-and-Resolution Architecture

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to articulate, in operational form, the composition relationship between conflict-as-first-class (A1.03) and directed selection (B1.14) — the resolution pair that together form the conflict visibility-and-resolution architecture — so that downstream work can adopt, extend, or argue against the pair as a composed unit without ambiguity.

Abstract

The Coordination Knowledge Substrate (CKS) pattern commits, in Paper 1, to registering conflicts as first-class substrate objects with their own identity and provenance rather than silently resolving them. It commits, in Paper 2, to directed selection as the governed mechanism by which humans authorize DNA modification to improve substrate specifications. These two commitments are individually defensible; this note formalizes that they are also architecturally dependent on each other in a specific way that makes neither fully functional without the other. A1.03 conflict-as-first-class provides the conflict targets that make directed selection informed rather than arbitrary; B1.14 directed selection provides the resolution pathway that makes conflict preservation meaningful rather than merely deferring conflicts indefinitely. Together they describe a conflict governance lifecycle — detection, registration, preservation, governance review, resolution decision, directed selection event, registry update — that requires both commitments to complete. The note states the pair identification, each enabling direction, the mutual dependency, the conflict lifecycle the pair describes, the co-presence requirement, and the operational detection criteria for whether a given system instantiates the pair.

1. Pair Identification

Pair: A1.03 (conflict-as-first-class) + B1.14 (directed selection).

Relationship type: Resolution pair.

A1.03 names the architectural commitment to treating conflicts in substrate specifications not as defects to be silently corrected, but as first-class substrate artifacts with their own identity, provenance, and addressability. Under A1.03, a conflict encountered in a DNA specification — two behavior substrates specifying incompatible behavior for the same input type, a content-domain boundary overlap between peer entities, a composition incompatibility between two cells' specifications — is registered in a conflict registry, preserved as live substrate state, and marked as requiring governance attention. A1.03 is a registration and preservation mechanism. It is not a resolution mechanism. It explicitly does not resolve the conflict.

B1.14 names the architectural commitment to directed selection as the primary governed mechanism for modifying DNA specifications. Under B1.14, human governors holding appropriate authority can authorize modifications to DNA-layer content — orchestration substrates, behavior substrates, content-domain boundary definitions, composition specifications — as a deliberate governance act with defined scope, recorded provenance, and traceable effect. B1.14 is a resolution mechanism. It is the governed pathway through which conflicts in DNA specifications are addressed.

The pair is named “resolution pair” because A1.03 and B1.14 address complementary halves of one architectural problem: what to do with conflicts that arise in substrate specifications. A1.03 answers the visibility half — conflicts must be registered, preserved, and made governable. B1.14 answers the resolution half — governance has a mechanism for actually addressing those conflicts through authorized DNA modification. Neither half alone answers both questions.

2. A1.03 Requires B1.14 for Resolution

Conflict-as-first-class per A1.03 is a registration and preservation commitment, not a resolution commitment. This is the commitment's essential character: A1.03 explicitly rejects silent resolution. It says that when a conflict is encountered, the system does not quietly pick one interpretation over another, does not invoke an implicit priority ordering, does not pass the conflict to an LLM for silent disambiguation. The conflict is registered as substrate content, marked with provenance, and preserved as a governance signal demanding attention.

The preservation is purposeful. A1.03 treats the conflict as evidence about the substrate's specification — evidence that two rules or boundaries are incompatible in ways governance has not yet addressed. Destroying that evidence through silent resolution destroys the governance signal. The point of preservation is to ensure that the people with authority over the substrate know the conflict exists and can act on it.

But preservation without a resolution mechanism produces only accumulation. If A1.03 registers every conflict and B1.14 does not exist as the resolution pathway, then every conflict enters the registry and none leave. The conflict backlog grows indefinitely. Governance sees the conflicts — A1.03 ensures visibility — but has no governed mechanism for addressing them. The conflicts become chronic, the registry becomes noise rather than signal, and the preservation commitment loses its purpose.

Three classes of DNA-level conflict are the primary targets for B1.14 resolution:

Conflicting DNA rules. When two behavior substrates specify incompatible behavior for the same input type, A1.03 registers the conflict. Directed selection per B1.14 resolves it: governance authorizes modification of one or both behavior substrates to eliminate the contradiction, with the modification recorded as a directed selection event and the conflict registry updated to reflect resolution.

Content-domain boundary conflicts. When content-domain boundaries overlap or conflict between peer entities — two cells claiming authority over the same input class — A1.03 registers the boundary conflict. Directed selection per B1.14 resolves it through boundary adjustment: one or both boundary definitions are modified under governance authority to eliminate the overlap.

Composition incompatibilities. When content-domain incompatibility makes two cells structurally incompatible for composition, A1.03 registers the composition conflict. Directed selection per B1.14 resolves it by modifying the content-domain specifications that produced the incompatibility.

In each case, the resolution mechanism is not automatic, not LLM-initiated, and not a cell-level rule application. It is a governance act: human review of the conflict registry, a decision about how to resolve, and an authorized DNA modification event that addresses the conflict at the specification level where it arose.

3. B1.14 Requires A1.03 for Informed Direction

Directed selection per B1.14 modifies DNA specifications under governance authority. The modification is purposeful — governance directs it toward a defined improvement. But purposefulness in the selection mechanism does not by itself ensure that the selection targets the places where conflicts exist. Without A1.03 conflict registration, conflicts in DNA specifications may be silently resolved at the cell level — one rule winning over another through implicit priority, an LLM disambiguation resolving a contradiction in context without recording it — without ever surfacing to governance as items requiring attention.

Silent resolution is the absence of A1.03. In a system without A1.03, conflicts do not accumulate in a visible registry; they disappear, handled locally, leaving no trace that they ever existed. Governance may run directed selection events to improve DNA specifications, but those events are aimed at improvements governance can see. Conflicts that were silently resolved are invisible to governance

— there is no registry pointing to them, no provenance recording that they occurred, no governance signal demanding attention. Directed selection in this environment may improve the substrate in many ways while leaving conflict-producing structural incompatibilities unaddressed because governance does not know they exist.

A1.03 changes this. With A1.03 in place, conflicts enter a conflict registry with provenance. Governance reviews the registry and can see which conflicts exist, where they arose, and what specification relationships produced them. Directed selection events can be targeted at the conflicts the registry identifies. The selection is informed: governance is directing modification at the substrate structures where conflicts are registered, with the conflict registry as the map of what needs attention.

The enabling direction from A1.03 to B1.14 is therefore an information-provision relationship. A1.03 provides the conflict visibility that makes directed selection conflict-resolving rather than merely improvement-seeking. Without A1.03, directed selection operates on a substrate whose conflict structure is partially or wholly invisible. With A1.03, directed selection has a map.

4. Mutual Dependency

The enabling directions named in §2 and §3 together constitute a mutual dependency: A1.03 and B1.14 each require the other to be architecturally coherent in their own terms.

A1.03 provides B1.14 with conflict targets. The conflict registry produced by A1.03 is the primary input that makes directed selection events conflict-resolving rather than generically improvement-oriented. Governance reviews the registry, identifies conflicts requiring resolution, and authorizes directed selection events specifically aimed at the conflicting specifications. The A1.03 registry is not the only source of directed selection targets — governance may also initiate DNA modifications for reasons unrelated to registered conflicts — but it is the source that makes directed selection systematically responsive to the conflict structure of the substrate.

B1.14 provides A1.03 with a resolution pathway. A1.03's preservation principle — don't silently resolve — is architecturally coherent only when a governed resolution pathway exists. The point of preserving conflicts is to make them governable so that governance can address them. If no governed pathway for addressing them existed, the preservation principle would produce only accumulating irresolvable conflicts. B1.14 provides the pathway that makes the principle work as designed: conflicts are preserved because they can be addressed, and they are addressed through directed selection.

This mutual dependency is asymmetric in its directionality but symmetric in its necessity. A1.03 without B1.14 produces a functional visibility commitment coupled to a non-functional resolution commitment: governance sees conflicts it cannot address through a governed mechanism. B1.14 without A1.03 produces a functional resolution mechanism operating on a substrate whose conflict structure is invisible: directed selection improves DNA without systematically reaching the conflicts that governance cannot see. Full co-presence produces the architecture both commitments require to

be complete.

5. The Conflict Governance Lifecycle

The pair together describes a conflict governance lifecycle. Each step in the lifecycle is claimed by one or both commitments; the lifecycle cannot complete if either commitment is absent.

Step 1 — Detection. A conflict is detected in a DNA specification: a behavior substrate contradiction, a content-domain boundary overlap, a composition incompatibility. Detection may occur during cell execution, during aspect-level review, or during governance-initiated inspection of substrate specifications.

Step 2 — Registration per A1.03. The conflict is registered as a first-class substrate artifact in the entity's conflict registry. Registration includes provenance: what specifications conflict, where they arose, when the conflict was detected, and what governance scope the conflict falls within. The conflict is not silently resolved at this step.

Step 3 — Preservation. The conflict persists as live substrate state. Both conflicting specifications remain in place; neither is automatically retired. The conflict registry entry remains active and addressable.

Step 4 — Governance review. Human governors with appropriate authority review the conflict registry. The review may be periodic (regularly scheduled) or triggered (by a registered conflict flagged for priority attention). Governance sees the conflict, understands the specification relationship that produced it, and can assess the resolution options.

Step 5 — Resolution decision. Governance decides how to resolve the conflict through directed selection: modify one conflicting specification, modify both, adjust a boundary definition, update a composition specification. The decision is a governance act, not a cell-level rule application or an LLM-initiated modification.

Step 6 — Directed selection event per B1.14. The authorized DNA modification is executed: the identified specification is modified under governance authority, with the modification recorded as a directed selection event with its own provenance. The modification may itself create new conflicts with other existing specifications — directed selection events have downstream effects that A1.03 will register if they produce new conflicts.

Step 7 — Conflict registry update. The conflict registry entry is marked resolved, with a reference to the directed selection event that addressed it. The resolution is itself substrate content with provenance: downstream inspection can trace how the conflict was addressed and by what governance authority.

This lifecycle applies recursively: at cell scope, aspect scope, and Self scope, conflict registration and directed selection resolution operate through the same lifecycle with level-appropriate DNA

modification. Multi-level simultaneous evolution means the lifecycle may be running at multiple levels concurrently under different human governors with different authority scopes.

6. Co-Presence Requirement

The co-presence requirement names what breaks when each commitment is present without the other.

A1.03 without B1.14. Conflict registration without a governed resolution mechanism. The conflict registry grows — every new conflict enters it — but nothing leaves. Governance reviews the registry and sees an accumulating backlog with no governed mechanism for addressing it. The conflicts become chronic; the registry becomes an inventory of permanent irresolvable incompatibilities rather than a governance signal driving improvement. Over time, the registry loses its function as a signal: if nothing in the registry can be addressed, the registry's content ceases to inform governance decisions. The preservation commitment drifts toward Specification Integrity Collapse as the backlog of unresolved conflicts grows without bound.

B1.14 without A1.03. Directed selection without conflict visibility. DNA modifications occur through a governed mechanism, but the modifications are not systematically aimed at conflicts because conflicts are not registered. Cell-level resolution handles contradictions silently — one specification wins, the conflict disappears without trace. Governance runs directed selection events that improve DNA in visible ways while leaving the substrate's invisible conflict structure intact. The result is a substrate that improves along the axes governance can see while accumulating structural incompatibilities along the axes governance cannot see. This produces Governance Theater in the conflict-resolution dimension: the appearance of governed improvement without the substance of governed conflict resolution.

Full co-presence. Conflicts are registered per A1.03 with provenance, making them visible governance targets. Directed selection events per B1.14 address registered conflicts through authorized DNA modification, closing the lifecycle at each conflict. The conflict registry grows when new conflicts arise and shrinks when directed selection events address existing ones. Governance has both the visibility to know which conflicts exist and the mechanism to address them through governed substrate modification.

7. Operational Detection

Three detection criteria indicate whether a system instantiates the A1.03 + B1.14 resolution pair as a composed unit.

A1.03 conflict registry integrity. Are conflicts in DNA specifications registered with provenance rather than silently resolved? Can a human governor inspect the conflict registry at any time and see which conflicts are active, when they were registered, and what specification relationships produced them? If conflicts are handled silently — one specification winning over another through implicit priority, an LLM resolving the contradiction in context without recording it — the A1.03

commitment is absent regardless of whether a conflict registry nominally exists.

B1.14 directed selection targeting. Do directed selection events reference the conflict registry? When governance authorizes a DNA modification, is the modification traceable to a conflict registry entry or to an explicit governance decision about substrate improvement? If directed selection events are recorded with provenance that does not reference the conflict registry, the connection between A1.03 visibility and B1.14 resolution is absent even if both mechanisms nominally exist in the same system.

Conflict resolution rate. Is the conflict registry backlog being addressed through directed selection, or is it growing without bound? A conflict registry that only accumulates — that grows with each new conflict registered but never decreases as conflicts are resolved — is the signature of A1.03 operating without B1.14 as its resolution pathway. A directed selection program that only improves DNA without ever resolving items in the conflict registry is the signature of B1.14 operating without A1.03 as its targeting map. Healthy co-presence produces a conflict registry that functions as a living governance instrument: entries arrive when conflicts are detected, entries are resolved when directed selection addresses them, and the registry at any time reflects the current state of the substrate’s specification conflicts.

A system that fails any of these three criteria may be useful, and may implement each commitment nominally, but does not instantiate the resolution pair as a composed architecture. The pair’s contribution is not the sum of its parts; it is the lifecycle the two commitments describe together.

8. Conclusion

Conflict-as-first-class per A1.03 and directed selection per B1.14 are individually defensible architectural commitments. This note formalizes that they are also architecturally dependent on each other in a way that makes the pair’s contribution distinct from either commitment alone. A1.03 provides the conflict visibility that makes directed selection informed; B1.14 provides the resolution pathway that makes conflict preservation meaningful. Together they describe a conflict governance lifecycle — from detection through registration through governance review through directed selection event through registry update — that requires both commitments to complete. A system that instantiates both commitments in operational connection, with directed selection events targeting the conflict registry and the conflict registry receiving updates when directed selection resolves entries, has instantiated the conflict visibility-and-resolution architecture. A system that instantiates either commitment alone has half of the architecture and none of the lifecycle.

Subsequent work that adopts the CKS pattern and needs to reason about conflict governance should engage both A1.03 and B1.14 as a pair. Subsequent work that argues against the pair — that proposes a different conflict governance architecture — should address both the visibility commitment and the resolution mechanism, and should name what replaces the lifecycle the pair describes.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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